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Capstone Part 2:

Part 6:

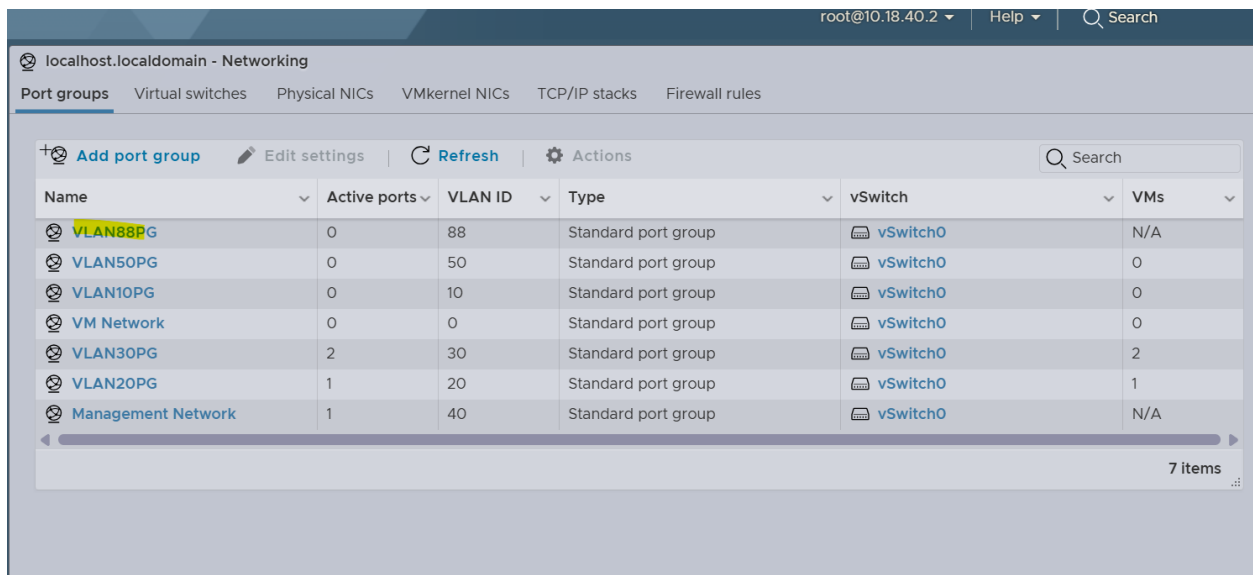
A new vlan named “DMZ” was created with an id of 88:

```
Enter configuration commands, one per line. End with CNTL/Z.  
LAYER3SWITCH(config)#vlan 88  
LAYER3SWITCH(config-vlan)#name DMZ  
LAYER3SWITCH(config-vlan)#exit  
LAYER3SWITCH(config)#
```

Interface **REDACTED** was configured for the DMZ.

```
Enter configuration commands, one per line. End with CNTL/Z.  
LAYER3SWITCH(config)#interface REDACTED  
LAYER3SWITCH(config-if)#switchport trunk allowed vlan add 88  
LAYER3SWITCH(config-if)#exit  
LAYER3SWITCH(config)#
```

I then created the port group in VMWare ESXi:



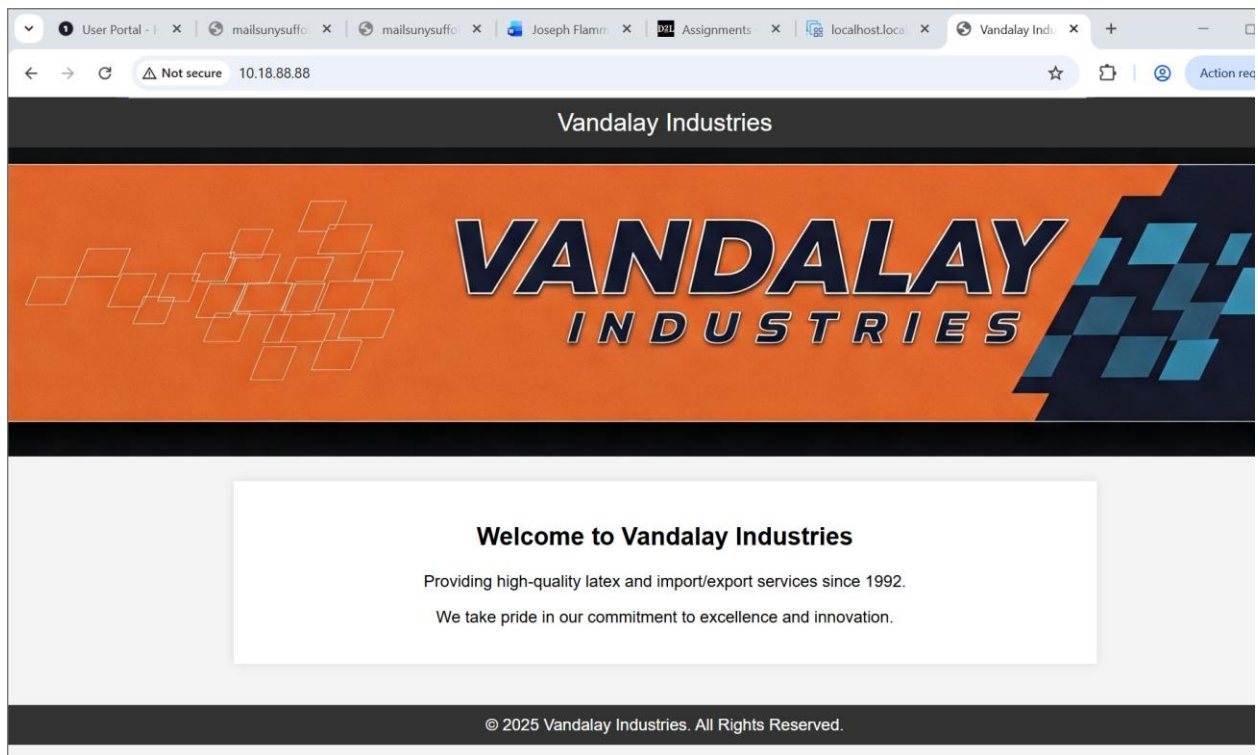
The screenshot shows the VMware ESXi Networking interface. The 'Port groups' tab is selected. A table lists the configured port groups. The 'VLAN88PG' port group is highlighted in yellow.

Name	Active ports	VLAN ID	Type	vSwitch	VMs
VLAN88PG	0	88	Standard port group	vSwitch0	N/A
VLAN50PG	0	50	Standard port group	vSwitch0	0
VLAN10PG	0	10	Standard port group	vSwitch0	0
VM Network	0	0	Standard port group	vSwitch0	0
VLAN30PG	2	30	Standard port group	vSwitch0	2
VLAN20PG	1	20	Standard port group	vSwitch0	1
Management Network	1	40	Standard port group	vSwitch0	N/A

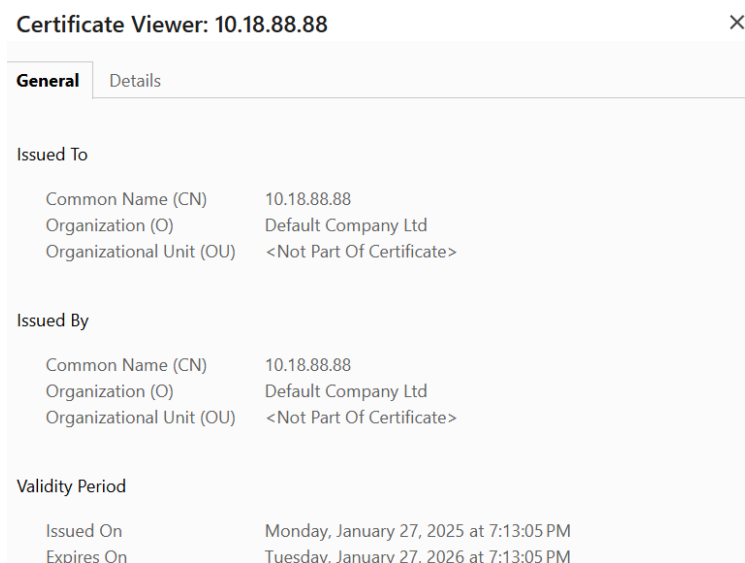
7 items

From here, I used the supplied VA web server and named it “DMZ-WEBSERVER”. This VM should be attached to VLAN 88.

Trying to access the VM via a browser with the IP now works successfully:

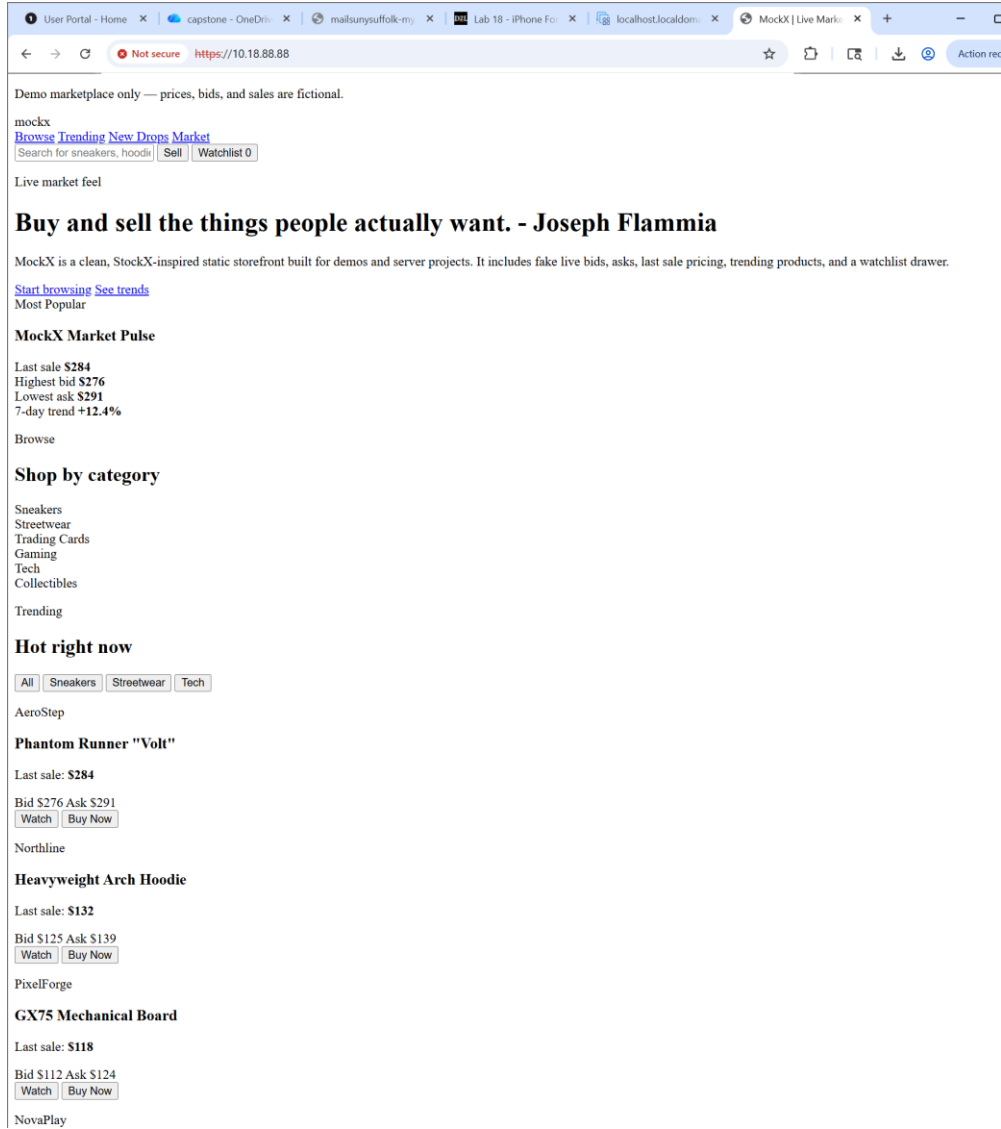


A self-signed certificate was created as shown below.



To edit the webpage, navigate to the “/var/www/html/” directory and enter “nano index.html”. As shown below, I added in my name next to the header of the site:

A restart to the httpd service is required to apply the changes. After entering “systemctl restart httpd” in the console, the webpage will appear as shown below:



```

LAYER3SWITCH(config)#router ospf 1
LAYER3SWITCH(config-router)#router
LAYER3SWITCH(config-router)#router-id 1.1.1.1
LAYER3SWITCH(config-router)#network 10.18.10.0 0.0.0.255 area 0
LAYER3SWITCH(config-router)#network 10.18.20.0 0.0.0.255 area 0
LAYER3SWITCH(config-router)#network 10.18.30.0 0.0.0.255 area 0
LAYER3SWITCH(config-router)#network 10.18.40.0 0.0.0.255 area 0
LAYER3SWITCH(config-router)#network 10.18.50.0 0.0.0.255 area 0
LAYER3SWITCH(config-router)#network 10.18.88.0 0.0.0.255 area 0
LAYER3SWITCH(config-router)#net REDACTED 0.0.0.255 area 0
LAYER3SWITCH(config-router)#exit
LAYER3SWITCH(config-router)#

```

For ACLs, an extended ACL was created that fulfilled two requirements. These being the TS environment being able to access our network, and then other students being able to access our DMZ page, but no other internal networks.

The ACL was applied on REDACTED and applied "In".

```
LAYER3SWITCH(config-ext-nacl)#inte REDACTED
LAYER3SWITCH(config-if)#ip acc
LAYER3SWITCH(config-if)#ip acce
LAYER3SWITCH(config-if)#ip access-group OUTSIDE-IN in
LAYER3SWITCH(config-if)#exit
```

Part 7:

Task 2 – Nagios:

Next, we will create a Linux VM that will run Nagios and monitor our network.

First, I went through the basic steps of installing Nagios. I did this all on a CentOS VM:

The screenshot shows the Nagios Core Services Platform web interface in a browser. The browser address bar shows the URL `http://192.168.142.129/nagios/`. The interface has a dark theme and a sidebar on the left with navigation links. The main content area displays a dashboard with various monitoring widgets.

Navigation Sidebar:

- General
 - Home
 - Documentation
- Current Status
- Tactical Overview
 - Map
 - Hosts
 - Services
 - Host Groups
 - Summary
 - Grid
 - Service Groups
 - Summary
 - Grid
- Problems
 - Services (Unhandled)
 - Hosts (Unhandled)
 - Network Outages
- Quick Search:
- Reports
 - Availability
 - Trends
 - Alerts
 - History
 - Summary
 - Histogram
 - Notifications
 - Event Log
- System
 - Comments
 - Downtime
 - Process Info
 - Performance Info
 - Scheduling Queue
 - Configuration

Main Dashboard:

- Version 4.5.9, December 19, 2024
- Daemon running with PID 1290
- Check for updates button
- Search Nagios: (input field)
- Changing button
- System Status widget: A grid of green and red squares representing host and service status.
- Latest Alerts widget: A table showing alerts with columns for Source, Location, Alert, and Status.
- Server Statistics widget: A table showing metrics like Load, CPU, Memory, and Disk.
- Service Grid widget: A grid showing the status of various services.
- Administrative Tasks widget: A section with buttons for tasks like 'Configure Nagios', 'Configure Nagios', 'Configure Nagios', and 'Configure Nagios'.

Footer:

- Meet Nagios Core Services Platform
- The next generation of Open Source powered monitoring with advanced dashboards, monitoring wizards, and much more!
- Learn More button
- Newsletter Sign-Up link

Now that nagios is installed, we will have to add the hosts in our network for monitoring. To do this, I edited the “etc/nagios/options/host.cfg” file and added the following hosts with the proper addressing and info:

```
Open ▾ + • define host use linux Draft

define host {
    use                linux-server ;
    host_name          DMZ-Web      ;
    address             10.18.88.88 ;
}

define host {
    use                linux-server ;
    host_name          WINDOWSSERVER2025 ;
    address             10.18.30.3 ;
}

define host {
    use                linux-server ;
    host_name          DC2WINSERV25 ;
    address             10.10.30.5 ;
}

?
```

For the DMZ webserver, I added multiple additional items to be monitored in the “services.cfg” file:

```
+ root@localhost:~ - nano /etc/nagios/objects/services.cfg

GNU nano 8.1 /etc/nagios/objects/services.cfg
define service {
    use                generic-service
    host_name          DMZ-WEBSEVER
    service_description PING
    check_command       check_ping!100.0,20%!500.0,60%
}

define service {
    use                generic-service
    host_name          DMZ-WEBSEVER
    service_description HTTP
    check_command       check_http
}

define service {
    use                generic-service
    host_name          DMZ-WEBSEVER
    service_description HTTPS
    check_command       check_http!-S
}

[ Read 27 lines ]
^G Help      ^O Write Out ^F Where Is  ^K Cut       ^C Location
^X Exit      ^R Read File ^\ Replace  ^U Paste     ^_ Justify   ^_ Go To Line
```

After these configurations, I set a manual IP on the system to one in the VLAN 30 subnet, restarted the nagios system, and then prepared to transfer it onto our main network.

Wired Cancel Apply

Details Identity **IPv4** IPv6 Security

IPv4 Method

☐ Automatic (DHCP) ☐ Link-Local Only

☒ Manual ☐ Disable

☐ Shared to other computers

Addresses

Address	Netmask	Gateway
10.18.30.20	255.255.255.0	10.18.30.1

DNS Automatic ☒

10.18.30.3

Separate IP addresses with commas

Routes Automatic ☒

Address	Netmask	Gateway	Metric

☐ Use this connection only for resources on its network

After adding the VM to the network, we can now see the VMs on the Nagios webpage:

Feb 16 7:12 PM

Nagios: 10.18.30.20 x +

Not Secure http://10.18.30.20/nagios/

Nagios

General

- Home
- Documentation
- Current Status**
- Tactical Overview
- Map
- Hosts
- Services
- Host Groups
 - Summary
 - Grid
- Service Groups
 - Summary
 - Grid
- Problems
 - Services (Unhandled)
 - Hosts (Unhandled)
 - Network Outages
- Quick Search:
- Reports
 - Availability
 - Trends

Current Network Status

Last Updated: Mon Feb 16 19:12:32 EST 2026
Updated every 90 seconds
Nagios® Core™ 4.5.9 - <https://nagios.org>
Logged in as nagiosadmin

Host Status Totals

Up	Down	Unreachable	Pending
1	0	0	3

All Problems All Types

All Problems	All Types
0	4

Service Status Totals

Ok	Warning	Unknown	Critical	Pending
7	1	0	0	0

All Problems All Types

All Problems	All Types
1	8

Host Status Details For All Host Groups

Limit Results: 100

Host	Status	Last Check	Duration	Status Information
DC2WINSRV25	PENDING	N/A	0d 0h 0m 7s+	Host check scheduled for Mon Feb 16 19:12:25 EST 2026
DM2-WEBSEVER	PENDING	N/A	0d 0h 0m 7s+	Host check scheduled for Mon Feb 16 19:13:40 EST 2026
WINDOWSSEVER2025	PENDING	N/A	0d 0h 0m 7s+	Host check scheduled for Mon Feb 16 19:14:55 EST 2026
localhost	UP	03-30-2026 12:08:48	???	PING OK - Packet loss = 0%, RTA = 0.04 ms

Results 1 - 4 of 4 Matching Hosts

On the services tab, we can see the additional services we listed for the DMZ site, which states that there are critical errors with HTTP and HTTPS:

Current Network Status

Last Updated: Mon Feb 16 19:29:45 EST 2026
 Updated every 90 seconds
 Nagios® Core™ 4.5.9 - www.nagios.org
 Logged in as nagiosadmin

[View History For all hosts](#)
[View Notifications For All Hosts](#)
[View Host Status Detail For All Hosts](#)

Host Status Totals

Up	Down	Unreachable	Pending
3	1	0	0

[All Problems](#)
[All Types](#)

1	4
---	---

Service Status Totals

Ok	Warning	Unknown	Critical	Pending
8	1	0	2	0

[All Problems](#)
[All Types](#)

3	11
---	----

Service Status Details For All Hosts

Limit Results: 100

Host	Service	Status	Last Check	Duration	Attempt	Status Information
DMZ-WEBSEVER	HTTP	CRITICAL	02-16-2026 19:29:15	0d 0h 0m 30s	1/3	connect to address 10.18.88.88 and port 80: Connection refused
	HTTPS	CRITICAL	02-16-2026 19:29:33	0d 0h 0m 12s	1/3	connect to address 10.18.88.88 and port 443: Connection refused
	PING	OK	02-16-2026 19:29:18	0d 0h 2m 48s+	1/3	PING OK - Packet loss = 0%, RTA = 0.56 ms
localhost	Current Load	OK	02-16-2026 19:20:42	???	1/4	OK - load average: 0.05, 0.25, 0.27
	Current Users	OK	02-16-2026 19:21:20	???	1/4	USERS OK - 1 users currently logged in
	HTTP	WARNING	02-16-2026 19:21:57	???	4/4	HTTP WARNING: HTTP/1.1 403 Forbidden - 1415363 bytes in 0.004 second response time
	PING	OK	02-16-2026 19:27:35	???	1/4	PING OK - Packet loss = 0%, RTA = 0.07 ms
	Root Partition	OK	02-16-2026 19:21:45	???	1/4	DISK OK - free space: / 23394 MiB (81.82% inode=99%):
	SSH	OK	02-16-2026 19:29:15	???	1/4	SSH OK - OpenSSH_9.9 (protocol 2.0)
	Swap Usage	OK	02-16-2026 19:29:27	???	1/4	SWAP OK - 67% free (1365 MB out of 2059 MB)
	Total Processes	OK	02-16-2026 19:25:05	???	1/4	PROCS OK: 198 processes with STATE = RSZDT

Results 1 - 11 of 11 Matching Services

Task 9 – RHEL Client:

Next, I will create a RHEL client and then configure it to authenticate to AD.

To start, I created a fresh RHEL client using a Rocky Linux ISO. The IP addressing was set as pictured below:

Cancel

Wired

Apply

Details

Identity

IPv4

IPv6

Security

IPv4 Method

☐ Automatic (DHCP)
 ☐ Link-Local Only

☒ Manual
 ☐ Disable

☐ Shared to other computers

Addresses

Address	Netmask	Gateway	
10.18.20.9	255.255.255.0	10.18.20.1	⊕
			⊖

DNS

Automatic ☒

10.18.30.3

Separate IP addresses with commas

Routes

Automatic ☒

Address	Netmask	Gateway	Metric

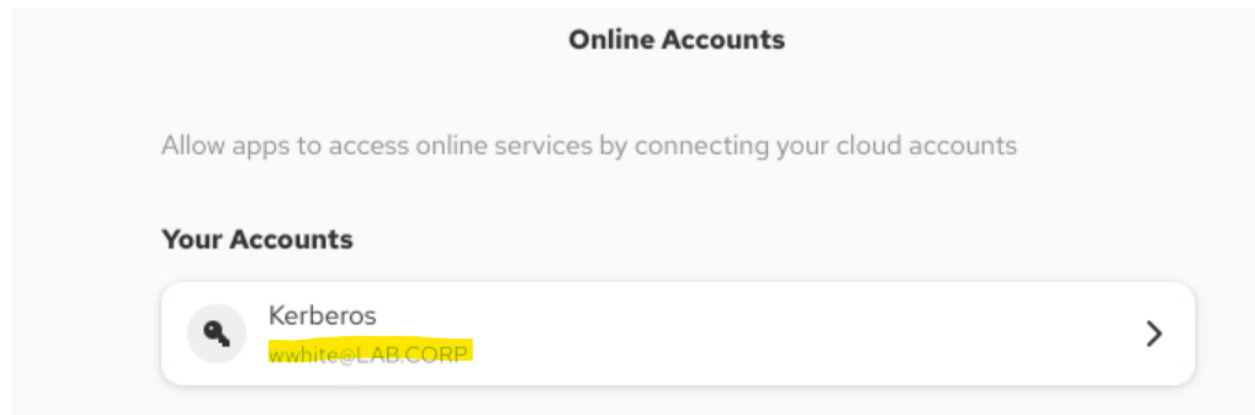
After transferring the VM to my network, I then installed the required dependencies and packages, and then entered the following command to join the domain:

```
root@rockyclient:/home/rocky# sudo realm join lab.corp -U LabAdmin
```

From here, I entered proper credentials to connect to the domain and then entered the following two commands to allow all domain users to log into Rocky and create home directories for them.

```
root@rockyclient:/home/rocky# sudo realm permit --all
root@rockyclient:/home/rocky# authselect select sssd with-mkhomedir --force
Backup stored at /var/lib/authselect/backups/2025-08-15-00-17-34.HkV8PD
Profile "sssd" was selected.
```

Attempting to sign on with a domain account is now successful as shown below:



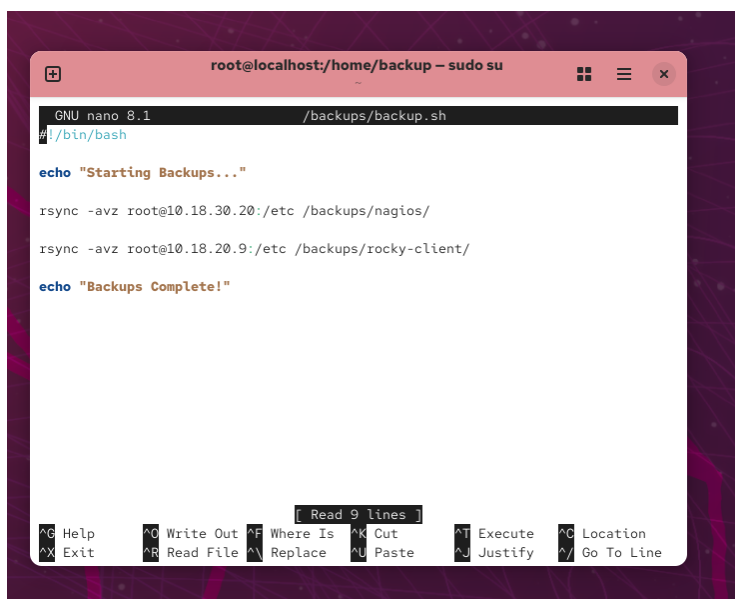
Task 8 – Backup Server:

Next, we will configure a backup server to contain scripts and software to back up our environment. This will be done on a Rocky Linux system.

To start, I entered to commands below to install “rsync”, and make a directory for the backups:

```
root@localhost:/home/backup# sudo dnf install rsync -y
Rocky Linux 10 - BaseOS                               488 kB/s | 16 MB      00:32
Rocky Linux 10 - AppStream                             3.6 MB/s | 2.3 MB    00:00
Rocky Linux 10 - Extras                               165 B/s | 6.0 kB     00:37
Package rsync-3.4.1-2.el10.x86_64 is already installed.
Dependencies resolved.
Nothing to do.
Complete!
root@localhost:/home/backup# sudo mkdir -p /backups
root@localhost:/home/backup# sudo chmod 777 /backups
root@localhost:/home/backup#
```

This fulfills the software requirement for the server. For the script, I created a simple script that will back up the Nagios server and the Rocky client, as pictured below:

A screenshot of a terminal window titled "root@localhost:/home/backup - sudo su". The terminal shows the execution of a script named "/backups/backup.sh" using GNU nano 8.1. The script's content is visible: it starts with a comment "#!/bin/bash", followed by an echo statement "Starting Backups...", then two rsync commands to back up the /etc directory of a Nagios server (root@10.18.30.20) and a Rocky client (root@10.18.20.9) into the /backups directory. The script concludes with an echo statement "Backups Complete!". The terminal also shows a status bar at the bottom with various keyboard shortcuts like ^G Help, ^X Exit, ^O Write Out, etc.

```
root@localhost:/home/backup - sudo su
GNU nano 8.1 /backups/backup.sh
#!/bin/bash

echo "Starting Backups..."

rsync -avz root@10.18.30.20:/etc /backups/nagios/

rsync -avz root@10.18.20.9:/etc /backups/rocky-client/

echo "Backups Complete!"

[ Read 9 lines ]
^G Help    ^O Write Out ^F Where Is ^K Cut      ^T Execute  ^C Location
^X Exit    ^R Read File ^\ Replace  ^P Paste    ^_ Justify  ^_ Go To Line
```

To make the script executable, I ran the command below:

```
root@localhost:/backups# chmod +x /backups/backup.sh
root@localhost:/backups#
```

Before transferring the VM, I set the addressing to the follow:

Cancel **Wired** Apply

Details Identity **IPv4** IPv6 Security

IPv4 Method

☐ Automatic (DHCP)
 ☐ Link-Local Only
 ☒ Manual
 ☐ Disable
 ☐ Shared to other computers

Addresses

Address	Netmask	Gateway	
10.18.30.25	255.255.255.0	10.18.30.1	⊗
			⊗

DNS Automatic ☒

10.18.30.3

Separate IP addresses with commas

Routes Automatic ☒

Address	Netmask	Gateway	Metric

To execute the backup, the command highlighted below can be entered, which backups the etc directory of the Nagios system:

```

root@localhost:/home/nagios - sudo su

etc/xdg/autostart/xdg-user-dirs.desktop
etc/xdg/menus/
etc/xdg/menus/applications.menu
etc/xdg/menus/gnome-applications.menu
etc/xdg/menus/applications-merged/
etc/xdg/systemd/
etc/xdg/systemd/user -> ../../systemd/user
etc/xml/
etc/xml/catalog
etc/xml/mallard/
etc/xml/mallard/catalog
etc/yum.repos.d/
etc/yum.repos.d/centos-addons.repo
etc/yum.repos.d/centos.repo
etc/yum.repos.d/epel-testing.repo
etc/yum.repos.d/epel.repo
etc/yum/
etc/yum/pluginconf.d -> ../dnf/plugins
etc/yum/protected.d -> ../dnf/protected.d
etc/yum/vars -> ../dnf/vars

sent 4,908,308 bytes received 30,352 bytes 1,411,045.71 bytes/sec
total size is 25,879,331 speedup is 5.24
root@localhost:/home/nagios# rsync -avz /etc backup@10.18.30.25:/backups/

```

To verify the backup was successful, on the backup server I ran the command “ls /backups”. As you can see, the etc directory was backed up successfully.

```

backup@localhost:~$ ls /backups
backup.sh etc
backup@localhost:~$

```

The same steps can be taken for the Rocky Client system.

To execute the script, on the backup server simply enter the command
“/backups/backups.sh”

```
root@localhost:/backups - sudo su

rsync: [sender] opendir "/etc/sudoers.d" failed: Permission denied (13)
rsync: [sender] send_files failed to open "/etc/security/opasswd": Permission denied (13)
rsync: [sender] send_files failed to open "/etc/ssh/ssh_host_ecdsa_key": Permission denied (13)
rsync: [sender] send_files failed to open "/etc/ssh/ssh_host_ed25519_key": Permission denied (13)
rsync: [sender] send_files failed to open "/etc/ssh/ssh_host_rsa_key": Permission denied (13)
rsync: [sender] send_files failed to open "/etc/ssh/sshd_config": Permission denied (13)
rsync: [sender] send_files failed to open "/etc/sysconfig/nftables.conf": Permission denied (13)
rsync: [sender] send_files failed to open "/etc/sysconfig/sshd": Permission denied (13)
rsync: [sender] send_files failed to open "/etc/wpa_supplicant/wpa_supplicant.conf": Permission denied (13)

sent 1,033 bytes  received 109,112 bytes  20,026.36 bytes/sec
total size is 25,191,866  speedup is 228.72
rsync error: some files/attrs were not transferred (see previous errors) (code 23) at main.c(1852) [generator=3.4.1]
Backups Complete!
root@localhost:/backups# /backups/backup.sh
```

Shown below are the backups of the Nagios and rocky client via the script:

```
root@localhost:/backups# ls
backup.sh  etc  nagios  rocky-client
root@localhost:/backups#
```

Output of etc directory from nagios system:

```

root@localhost:/backups/nagios# ls
etc
root@localhost:/backups/nagios# cd etc/
root@localhost:/backups/nagios/etc# ls
adjtime                gshadow                pnm2ppa.conf
aliases                gshadow-              polkit-1
alsa                  gss                   popt.d
alternatives           host.conf              printcap
anacrontab             hostname               profile
asound.conf            hosts                  profile.d
at.deny                hp                     protocols
audit                 httpd                  pulse
authselect             inittab                qemu-ga
avahi                  inputrc                ras
bash_completion.d     ipp-usb                rc.d
bashrc                 iproute2               rc.local
bindresvport.blacklist iscsi                  redhat-release
binfmt.d               issue                  request-key.conf
bluetooth              issue.d                request-key.d

```

Task 1 – Docker:

The next task will be task 1. For the docker server, I opted to use an Ubuntu CLI.

After installing docker and nginx, I opted to use the “Dashy” dashboard, a simple GUI interface site. Externally, the site runs over port 80 and internally runs over 8080.

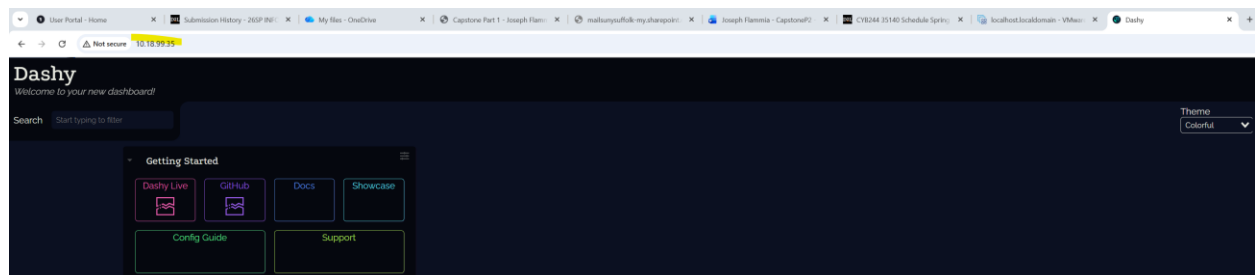
Docker Webserver Command:

```

root@dockerserver:/home/dockeruser# docker run -d -p 80:8080 --name webserver lissy93/dashy_

```

Site on Browser:



After transferring the VM to my network, I created a new VLAN to specifically serve the webserver:

+ Add port group - Webdev

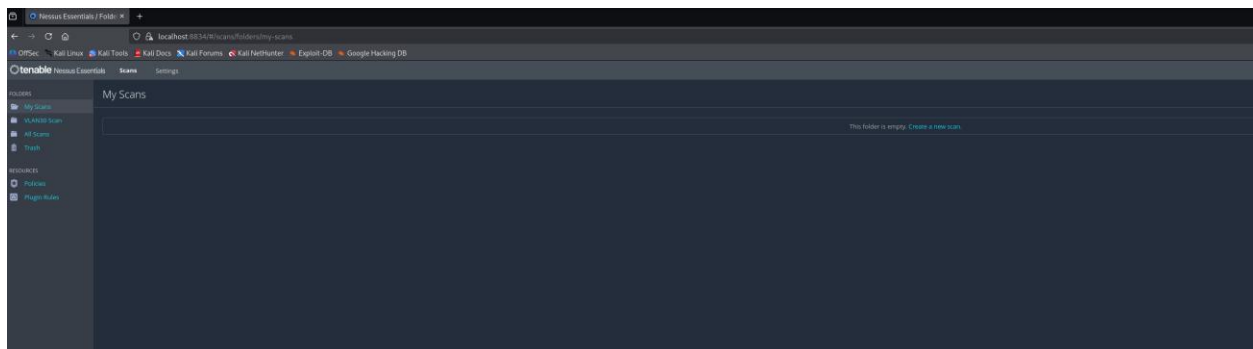
Name	Webdev
VLAN ID	99
Virtual switch	vSwitch0
> Security	Click to expand

CANCEL ADD

```
LAYER3SWITCH(config-vlan)#interface vlan 99
LAYER3SWITCH(config-if)#ip
LAYER3SWITCH(config-if)#ip ad
LAYER3SWITCH(config-if)#ip add
LAYER3SWITCH(config-if)#ip address 10.18.99.1 255.255.255.0
LAYER3SWITCH(config-if)#no shutdown
LAYER3SWITCH(config-if)#
```

Task 3 – Nessus:

For this system, I used Kali Linux, below is a screenshot of Nessus accessed via the web browser



The network I scanned was the “VLAN 30” network, since it hosts the servers which contain valuable information. Below is an output of the network scan.

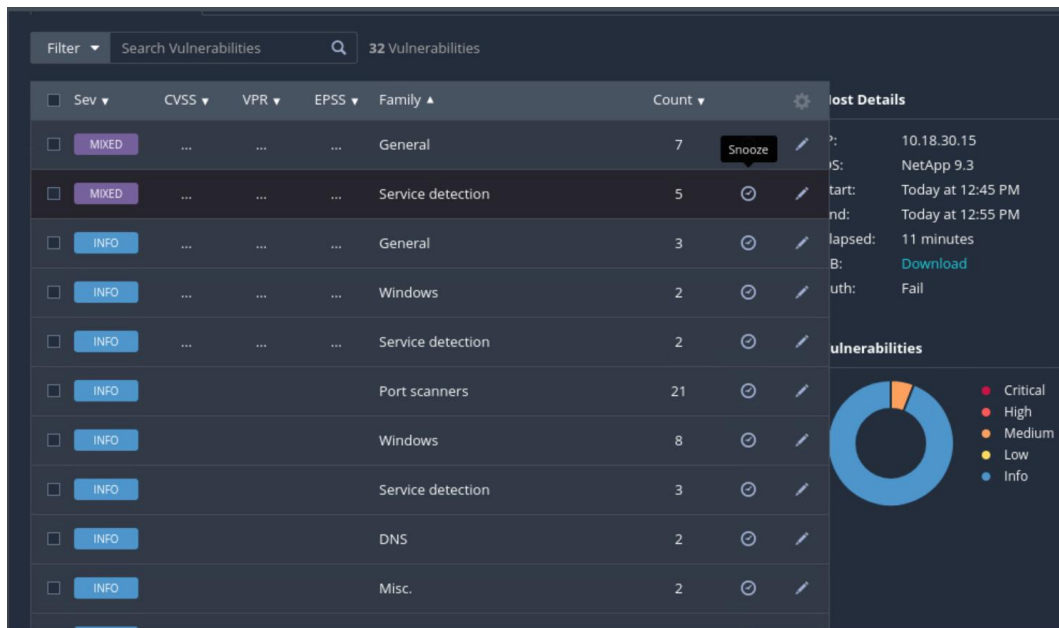


Hosts 5 Vulnerabilities 67 Remediations 1 Notes 7 History 1

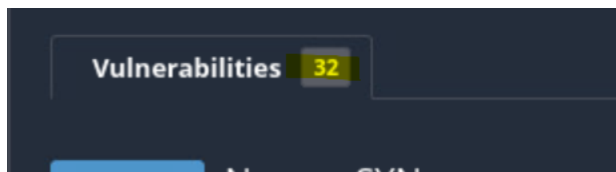
Filter Search Vulnerabilities 67 Vulnerabilities

Sev	CVSS	VPR	EPSS	Family	Count	
CRITICAL	9.8			Service detection	1	
MIXED	...	...	...	Windows	20	
MIXED	...	...	...	General	27	
MIXED	...	...	...	General	6	
MIXED	...	...	...	Windows	4	
MIXED	...	...	...	Misc.	2	
MIXED	...	...	...	Web Servers	13	
MIXED	...	...	...	Service detection	9	
MIXED	...	...	...	General	2	
MIXED	...	...	...	Windows	2	

Below is a screenshot of an in-depth scan run on DC2 (10.18.30.15). I chose this system since it had the most vulnerabilities.



The most interesting piece of information I found was not a specific vulnerability, but a piece of information. As shown below, I was able to see all ports that are open on the system which came out to 32:

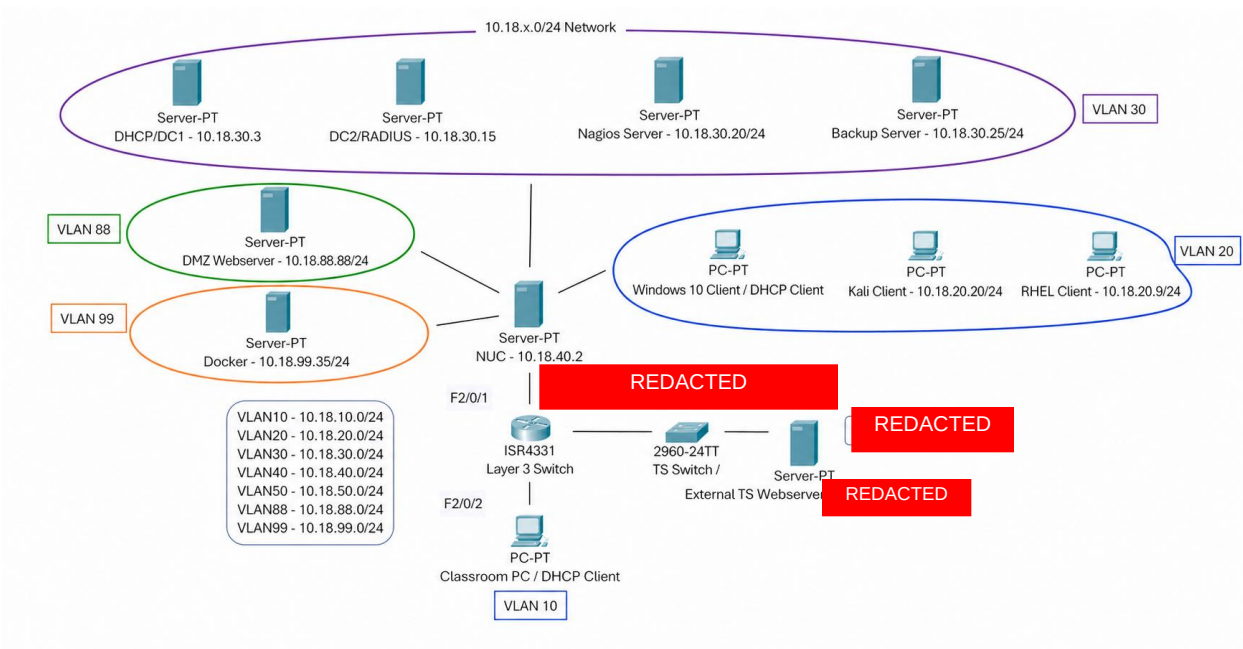


Port 22/tcp was found to be open	
To see debug logs, please visit individual host	
Port	Hosts
22 / tcp / ssh	10.18.30.15

Port 53/tcp was found to be open	
To see debug logs, please visit individual host	
Port	Hosts
53 / tcp / dns	10.18.30.15

Port 88/tcp was found to be open	
To see debug logs, please visit individual host	
Port	Hosts
88 / tcp	10.18.30.15

Below is a screenshot of the topology as of completing part 7:



Below are pings from the classroom PC to all the new devices add in parts 6 and 7:

```
C:\Users\flamj95>ping 10.18.30.20
Pinging 10.18.30.20 with 32 bytes of data:
Reply from 10.18.30.20: bytes=32 time=1ms TTL=63

Ping statistics for 10.18.30.20:
    Packets: Sent = 1, Received = 1, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 1ms, Maximum = 1ms, Average = 1ms
Control-C
^C
C:\Users\flamj95>ping 10.18.30.25
Pinging 10.18.30.25 with 32 bytes of data:
Reply from 10.18.30.25: bytes=32 time=1ms TTL=63

Ping statistics for 10.18.30.25:
    Packets: Sent = 1, Received = 1, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 1ms, Maximum = 1ms, Average = 1ms
Control-C
^C
C:\Users\flamj95>ping 10.18.30.35
Pinging 10.18.30.35 with 32 bytes of data:
Control-C
^C
C:\Users\flamj95>ping 10.18.99.35
Pinging 10.18.99.35 with 32 bytes of data:
Reply from 10.18.99.35: bytes=32 time=1ms TTL=63

Ping statistics for 10.18.99.35:
    Packets: Sent = 1, Received = 1, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 1ms, Maximum = 1ms, Average = 1ms
Control-C
^C
C:\Users\flamj95>ping 10.18.88.88
Pinging 10.18.88.88 with 32 bytes of data:
Reply from 10.18.88.88: bytes=32 time=1ms TTL=63

Ping statistics for 10.18.88.88:
    Packets: Sent = 1, Received = 1, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 1ms, Maximum = 1ms, Average = 1ms
Control-C
^C
C:\Users\flamj95>ping 10.18.20.9
Pinging 10.18.20.9 with 32 bytes of data:
Reply from 10.18.20.9: bytes=32 time=1ms TTL=63

Ping statistics for 10.18.20.9:
    Packets: Sent = 1, Received = 1, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 1ms, Maximum = 1ms, Average = 1ms
Control-C
^C
C:\Users\flamj95>ping 10.18.20.20
Pinging 10.18.20.20 with 32 bytes of data:
Reply from 10.18.20.20: bytes=32 time=1ms TTL=63

Ping statistics for 10.18.20.20:
    Packets: Sent = 1, Received = 1, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
```

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C:\Users\flamj95>ping 10.18.88.1 **VLAN 88**

Pinging 10.18.88.1 with 32 bytes of data:
Reply from 10.18.88.1: bytes=32 time=2ms TTL=255

Ping statistics for 10.18.88.1:
Packets: Sent = 1, Received = 1, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
Minimum = 2ms, Maximum = 2ms, Average = 2ms

Control-C

^C

C:\Users\flamj95>ping 10.18.99.1 **VLAN 99**

Pinging 10.18.99.1 with 32 bytes of data:
Reply from 10.18.99.1: bytes=32 time=1ms TTL=255

Ping statistics for 10.18.99.1:
Packets: Sent = 1, Received = 1, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
Minimum = 1ms, Maximum = 1ms, Average = 1ms

Control-C

^C